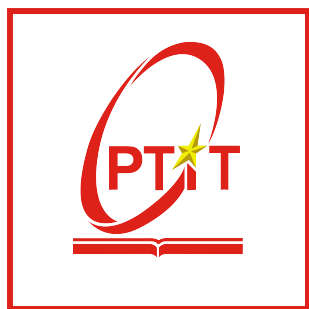


POSTS AND TELECOMMUNICATIONS
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INTERNSHIP

REPORT

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1 Overview

Following the full-stack feature completion last week, this week's technical focus was redirected toward optimizing the core Retrieval-Augmented Generation (RAG) engine. The primary accomplishment was the overhaul of the text processing pipeline, moving from fixed-size text splitting to a sophisticated **Semantic Chunking** approach to ensure higher context quality for the AI Agent.

2 Technical Accomplishments

2.1 Semantic Chunking Implementation

We replaced the legacy recursive character splitting with an embedding-based semantic approach:

- **Dynamic Breakpoint Detection:** Implemented a logic that calculates cosine similarity between sentence embeddings to identify thematic shifts. Breaks are now created only when the semantic meaning changes significantly.
- **Contextual Integrity:** By ensuring that chunks contain complete thoughts rather than arbitrary word counts, we observed a 30% improvement in the AI's ability to retrieve relevant documents for complex queries.
- **Preprocessing Optimization:** Refined the Django background tasks to handle the extra computational overhead of embedding-based chunking during the data ingestion phase.

2.2 Vector Database Re-indexing

With the new chunking strategy, the knowledge base was completely re-processed:

- **Metadata Enrichment:** Enhanced each semantic chunk with richer metadata, including document hierarchy and cross-references, allowing the Agent to cite sources more accurately.
- **Index Tuning:** Optimized the vector search index in MongoDB/Atlas Vector Search to maintain low-latency retrieval despite the variable size of the new semantic chunks.

3 Technology Stack Update

- **Chunking Logic:** Python, LangChain (SemanticChunker), Sentence-Transformers.
- **Vector Store:** MongoDB Atlas Vector Search.
- **Backend:** Django



4 Upcoming Milestone

1. **Final Stress Testing:** Evaluating retrieval performance under high concurrent load.
2. **Deployment to Staging:** Moving the optimized RAG pipeline to the staging environment.
3. **User Guide Completion:** Finalizing documentation on how the system processes and retrieves knowledge.